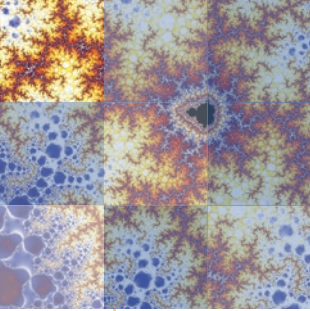




Solving Radical Equations

A **radical equation** has at least one radical expression with a variable in the radicand.

Exponential Property of Equality

If $a = b$ and n is a positive integer, then $a^n = b^n$.

Solving Radical Equations

Step 1 Isolate the radical on one side of the equation.

Step 2 Raise both sides of the equation to the same power to eliminate the radical.

Step 3 Solve the equivalent equation.

Step 4 Check each solution by substituting into the original equation.

The power of a power property, $(a^m)^n = a^{mn}$, makes it possible to use the exponential property of equality to solve radical equations.

► **Think About It** Raising to the n th power is the inverse operation of taking the n th root.

EXAMPLE 1

Solve and check.

A $\sqrt{x + 12} = 8$

SOLUTION

$$\sqrt{x + 12} = 8$$

$$(\sqrt{x + 12})^2 = 8^2 \quad \text{Square both sides of the equation.}$$

$$x + 12 = 64 \quad \text{Simplify.}$$

$$x = 52 \quad \text{Subtract 12 from both sides.}$$

CHECK

Substitute 52 for x .

$$\begin{aligned}\sqrt{52 + 12} &\stackrel{?}{=} 8 \\ \sqrt{64} &\stackrel{?}{=} 8 \\ 8 &= 8 \quad \checkmark\end{aligned}$$

The solution is $x = 52$.

B $-5\sqrt[3]{a - 1} = 25$

SOLUTION

$$-5\sqrt[3]{a - 1} = 25$$

$$\sqrt[3]{a - 1} = -5 \quad \text{Isolate the radical.}$$

$$(\sqrt[3]{a - 1})^3 = (-5)^3 \quad \text{Cube both sides of the equation.}$$

$$a - 1 = -125 \quad \text{Simplify.}$$

$$a = -124 \quad \text{Add 1 to both sides.}$$

CHECK

Substitute -124 for a .

$$\begin{aligned}-5\sqrt[3]{-124 - 1} &\stackrel{?}{=} 25 \\ -5\sqrt[3]{-125} &\stackrel{?}{=} 25 \\ -5(-5) &\stackrel{?}{=} 25 \\ 25 &= 25 \quad \checkmark\end{aligned}$$

The solution is $a = -124$. ■

► **Remember** The third root of a negative number is always negative.

Identifying an Extraneous Solution

Raising each side of an equation to a power can create extraneous solutions. Always check your solutions to be certain each one satisfies the original equation.

EXAMPLE 2

Solve $x + 1 = \sqrt{5x + 11}$.

SOLUTION

$$x + 1 = \sqrt{5x + 11}$$

$$(x + 1)^2 = (\sqrt{5x + 11})^2 \quad \text{Square both sides of the equation.}$$

$$x^2 + 2x + 1 = 5x + 11 \quad \text{Simplify.}$$

$$x^2 - 3x - 10 = 0 \quad \text{Set equal to 0.}$$

$$(x - 5)(x + 2) = 0 \quad \text{Factor.}$$

$$x = 5 \quad \text{or} \quad x = -2 \quad \text{Zero Product Property}$$

CHECK

Substitute 5 for x .

$$5 + 1 \stackrel{?}{=} \sqrt{5 \cdot 5 + 11}$$

$$6 \stackrel{?}{=} \sqrt{36}$$

$$6 = 6 \quad \checkmark$$

Substitute -2 for x .

$$-2 + 1 \stackrel{?}{=} \sqrt{5(-2) + 11}$$

$$-1 \neq \sqrt{1}$$

The solution $x = -2$ is extraneous. The only solution is $x = 5$. ■

► **Remember** An extraneous solution is an apparent solution that does not satisfy the original equation.

Solving a Radical Equation with Two Radicals

EXAMPLE 3

Solve $\sqrt{3m-2} = \sqrt{m+10}$.

SOLUTION

$$\sqrt{3m-2} = \sqrt{m+10}$$

$$(\sqrt{3m-2})^2 = (\sqrt{m+10})^2$$

Square both sides of the equation.

$$3m-2 = m+10$$

Simplify.

$$2m = 12$$

Subtraction and Addition Properties of Equality

$$m = 6$$

Division Property of Equality ■

When squaring an equation with radicals on both sides, clear all variables from radical expressions, isolate the radical, square both sides again, and simplify.

EXAMPLE 4

Solve $\sqrt{x+13} = \sqrt{12-x} - 1$.

SOLUTION

$$\sqrt{x+13} = \sqrt{12-x} - 1$$

$$(\sqrt{x+13})^2 = (\sqrt{12-x} - 1)^2$$

Square both sides of the equation.

$$x+13 = 12-x-2\sqrt{12-x}+1$$

Use the binomial square pattern on the right side.

$$2x = -2\sqrt{12-x}$$

Simplify.

$$-x = \sqrt{12-x}$$

Isolate the radical.

$$(-x)^2 = (\sqrt{12-x})^2$$

Square both sides of the equation.

$$x^2 = 12-x$$

Simplify.

$$x^2 + x - 12 = 0$$

Set equal to 0.

$$(x+4)(x-3) = 0$$

Factor.

$$x = -4 \quad \text{or} \quad x = 3$$

Zero Product Property

CHECK

Substitute -4 for x .

$$\begin{aligned}\sqrt{-4 + 13} &\stackrel{?}{=} \sqrt{12 - (-4)} - 1 \\ \sqrt{9} &\stackrel{?}{=} \sqrt{16} - 1 \\ 3 &= 3 \checkmark\end{aligned}$$

Substitute 3 for x .

$$\begin{aligned}\sqrt{3 + 13} &\stackrel{?}{=} \sqrt{12 - 3} - 1 \\ \sqrt{16} &\stackrel{?}{=} \sqrt{9} - 1 \\ 4 &\neq 2\end{aligned}$$

The solution $x = 3$ is extraneous. The only solution is $x = -4$. ■

Rearranging Formulas

You can rearrange a mathematical formula to solve for a particular value. Determine which variable you want to isolate and then solve the equation for that quantity.

EXAMPLE 5

Isaac Newton derived a formula to describe the speed a rocket must travel to escape the gravitational pull of a planet. This speed is called the escape velocity, or v_{esc} .

$v_{esc} = \sqrt{\frac{2GM}{r}}$, where G is the gravitational constant, M is the mass of the planet, and r is the radius of the planet. Solve the formula for r .

► **Remember** When solving a radical equation, square both sides of the equation to clear the radical.

SOLUTION

$$v_{esc} = \sqrt{\frac{2GM}{r}}$$

$$(v_{esc})^2 = \left(\sqrt{\frac{2GM}{r}} \right)^2$$

$$(v_{esc})^2 = \frac{2GM}{r}$$

$$r \bullet (v_{esc})^2 = \frac{2GM}{\cancel{r}} \bullet \cancel{r}$$

$$\frac{r \bullet \cancel{(v_{esc})^2}}{\cancel{(v_{esc})^2}} = \frac{2GM}{\cancel{(v_{esc})^2}^2}$$

$$r = \frac{2GM}{(v_{esc})^2}$$

Square both sides of the equation.

Simplify.

Multiply both sides by r .

Divide both sides of the equation by $(v_{esc})^2$.

Simplify. ■